

APEX GLOBAL BANK — Data Dictionary

Emerging Market Entrepreneurs Loan Product · 1,000 borrowers · 20 features + 1 target

Why this feature mix?

Emerging-market entrepreneurs typically have **thin traditional credit files**. Standard models that lean on credit score, income, and collateral systematically decline creditworthy borrowers from this segment. We balanced **10 traditional** and **10 behavioral** features so the model can detect creditworthiness through patterns of *behavior* — bill consistency, savings habits, community engagement, app activity, and network quality — which legacy scorecards ignore.

Traditional features (10)

Variable	Type	Description & rationale
Age	int	Borrower age (18–70). Proxy for life-stage financial stability.
Monthly_Income	float	Self-reported monthly income in USD (\$430–\$25,000). Lognormal spread typical of emerging markets.
Debt_to_Income	float	Ratio of existing debt to income (0.05–0.92). Strong traditional risk indicator.
Credit_Score	int	Bureau score (300–850). Industry standard; weak alone for thin-file borrowers but informative when combined.
Loan_Amount	float	Requested principal in USD (\$5,500–\$90,000). Higher principal = larger exposure on default.
Employment_Years	float	Tenure at current employer. Longer tenure → cash-flow predictability.
Dependants	int	Household dependants (0–7). Higher count strains disposable income.
Prior_Defaults	int	Past defaults on record (0–5). Strongest single traditional risk flag.
Collateral_Value	float	Value of pledged collateral in USD. Recovers principal on default.
Credit_Accounts	int	Number of active credit accounts. Thin-file proxy.

Behavioral features (10)

Variable	Type	Description & rationale
Utility_Latency	float	Avg days late on utility bills (0–45). Direct behavioral signal of payment discipline.
App_Session_Time	float	Minutes spent in the banking app per month. Proxy for financial attention.
Social_Sentiment	float	Sentiment score from anonymized social/community signals (–1 to +1). Reputation proxy.
Mobile_Data_Use	float	Mobile data usage (GB/month). Digital-economy participation.
Ecommerce_Freq	int	E-commerce purchases per month. Diversified spending = broader engagement.
Savings_Score	float	Composite savings discipline score (0–100). Indicates buffer capacity.
Community_Score	float	Reputation/standing in local community (0–100). Soft trust signal.
Night_Activity	float	Share of activity at night (0–1). High values can indicate irregular cash patterns.
Contact_Diversity	int	Distinct transaction/contact counterparties. Network-quality proxy.
Bill_Consistency	float	Consistency of bill payments (0–1). High = predictable, low = erratic.

Target variable & ethical notes

Loan_Status (binary): **0** = Default, **1** = Paid. Base default rate ≈ **20%** in the dataset. **Bias guardrails:** we exclude all protected attributes (gender, ethnicity, religion). The agent always surfaces mitigating factors and routes Grey-Zone cases to a human loan officer — humans hold the final decision. Every decision is logged with a full audit trail.